

Deep Reinforcement Learning for Cryptocurrency Portfolio Management: A Free-Energy Framework with Geometry-Based Transaction Costs and Efficiency Bounds

Ntebogang Dinah Moroke ^{1,*}  0000-0001-8545-1860

¹ Faculty of Economic and Management Sciences, North-West University, Mmabatho 2735, South Africa; ntebogang.moroke@nwu.ac.za

Abstract

This paper develops a deep reinforcement learning framework for cryptocurrency portfolio management in which transaction costs are derived from the Riemannian geometry of the underlying volatility model rather than assumed constant. A Proximal Policy Optimisation agent is trained on a reward function grounded in non-equilibrium thermodynamics: the free-energy Bellman equation, in which transaction costs are the geodesic slippage on the Fisher information manifold of a maximum-entropy Markov-switching GARCH model, and regime-transition costs are the Wasserstein-2 distance between the calm and turbulent return distributions. A thermodynamic Carnot bound on portfolio efficiency is established and empirically validated. Five hypotheses are tested across Bitcoin, Ethereum, Ripple, Litecoin, and Bitcoin Cash over January 2017 to March 2026. The geometric-cost agent achieves statistically superior Sharpe ratios relative to flat-fee baselines on four of five assets; portfolio turnover is reduced by 56 to 83 percent relative to signal-following; the thermodynamic friction point at which the agent prefers no-trade is asset-specific and ordered by turbulent half-life; a joint topological and geometric circuit breaker reduces Maximum Drawdown by 28 to 38 percent; and ablation confirms that every component of the observation vector contributes a statistically significant performance gain. The framework requires liquid cryptocurrency markets with validated parametric volatility models; transferability to other asset classes requires upstream recalibration.

Keywords: cryptocurrency portfolio optimisation; free-energy Bellman equation; geodesic transaction costs; proximal policy optimisation; reinforcement learning; thermodynamic efficiency; Wasserstein dissipation; topological circuit breakers

1. Introduction

The rapid expansion of digital asset markets has created both an opportunity and a methodological challenge for quantitative portfolio management. Cryptocurrency markets now account for over USD 2 trillion in aggregate market capitalisation, with daily trading volumes exceeding USD 100 billion across major exchanges (Liu et al. 2025). Unlike traditional equity or fixed income markets, cryptocurrency returns exhibit extreme fat tails, persistent volatility clustering, and abrupt distributional regime shifts that compress and extend over timescales ranging from hours to months (Cont 2001; Dyhrberg 2016). These structural features create a distinctive challenge: the statistical geometry of the market state changes so rapidly that any portfolio strategy calibrated on recent data may face dramatically different execution conditions by the time it reaches the order book. Standard

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mean- variance portfolio theory (Markowitz 1952) and its extensions (Sharpe 1966) were developed under distributional stationarity assumptions that cryptocurrency markets systematically violate. Reinforcement learning offers a natural framework for dynamic decision-making under non-stationarity (Jiang and Liang 2017; Moody and Saffell 1998), but existing RL portfolio implementations share a critical weakness: they treat transaction costs as a flat proportional fee (Hambly et al. 2023), ignoring the geometric reality that execution costs depend on the distributional state of the market.

This paper addresses that weakness directly. The core observation is that the cost of a portfolio rebalancing is not fixed: it depends on how much the trade displaces the market's conditional return distribution from its current state. When the distribution is in a stable, narrow-tailed calm regime, a small portfolio rebalancing causes minimal distributional displacement and incurs low execution costs. When the distribution has shifted into a wide, fat-tailed turbulent regime, the same rebalancing displaces the distribution substantially, incurring costs that can be 50% or more above any flat proportional estimate. Information geometry (Amari 2016; Rao 1945) provides the natural mathematical language for this observation: the geodesic distance on the Fisher information manifold of the underlying volatility model gives a state-dependent transaction cost that increases precisely when market conditions make trading expensive. In addition to the within-regime cost, crossing a regime boundary from calm to turbulent requires a minimum thermodynamic work payment, measured by the Wasserstein-2 distance between the two return distributions (Backhoff et al. 2020; Villani 2003). Both costs are integrated into a free-energy Bellman equation (Kappen 2005; Levine 2018), and a fundamental efficiency limit, a portfolio analogue of the Carnot bound, is derived from the Jarzynski equality (Jarzynski 1997).

To make the thermodynamic framework financially concrete from the outset, three correspondences are established throughout the paper and their economic content stated plainly. Geodesic slippage corresponds to state-dependent execution cost: the statistical distance the market must travel to absorb a trade. A high geodesic cost on a given day means that executing any trade pushes the return distribution far from its current calibration, and the market will price that displacement into the bid-ask spread and depth withdrawal. Wasserstein dissipation corresponds to the minimum thermodynamic work required to cross a regime boundary: a portfolio strategy that rebalances during a calm-to-turbulent transition is paying for the distributional transport, not merely the position change. The Carnot efficiency bound corresponds to the maximum risk-adjusted return per unit of Wasserstein dissipation: no learning-based agent, regardless of its architecture, can extract more risk-adjusted value from a given market than the entropy differential between regimes permits. These three quantities are computable from the same upstream statistical model and require no free parameters beyond what the volatility model already estimates.

This paper is Paper 3 of the Window-Orchestrated Wealth – Extended Wasserstein (WOW-E-W) quadrilogy, a four-paper research programme that progressively builds a unified thermodynamic architecture for cryptocurrency portfolio management. A self-contained summary of all pipeline equations is provided in Section 3.1 and the full computational pipeline is publicly available at https://github.com/ntebo40/IST03_Quadrilogy (Zenodo DOI: <https://doi.org/10.5281/zenodo.19072906>, CC-BY 4.0), enabling independent verification of every upstream input.

1.1. Contributions

This paper makes four distinct contributions to the intersection of reinforcement learning, information geometry, and thermodynamic portfolio theory.

Contribution 1: Free-energy Bellman equation with Wasserstein dissipation. Prior portfolio RL reward functions use KL divergence as the cost of policy change (Haarnoja

et al. 2018; Levine 2018). KL divergence is infinite when distributions have disjoint support, which is common during regime transitions where calm and turbulent returns have minimal overlap. This paper introduces a specialised form using Wasserstein-2 distance as the regime-transition cost, which remains finite and measures the actual earth-mover distance between distributional states. The resulting free-energy Bellman equation penalises trading precisely when execution geometry is most adverse, providing a principled basis for identifying optimal rebalancing timing.

Contribution 2: Geodesic transaction costs. All prior portfolio RL implementations use flat proportional fees (Hambly et al. 2023; Jiang and Liang 2017). This paper replaces the flat fee with the geodesic slippage S^* , defined as the Riemannian arc length on the Fisher information manifold of the MS-GARCH-MaxEnt parameter path (Section 3.2). The flat-fee assumption is a degenerate special case corresponding to an empirically incorrect assumption of zero sectional curvature. The asset-specific friction point analysis in Section 5.3 provides actionable fee thresholds for practitioners: it identifies when active management becomes thermodynamically inefficient for each asset.

Contribution 3: Thermodynamic Carnot bound. Theorem 1 establishes $\eta \leq 1 - H_{\text{turb}}/H_{\text{calm}}$, the first efficiency bound derived for a learning-based trading agent from thermodynamic first principles via the Jarzynski equality. The bound is empirically validated across five assets with Spearman rank correlation $\rho = 0.94$ between realised efficiency and turbulent half-life. The efficiency metric η provides a monitoring diagnostic: a sustained decline below 50 percent of the Carnot bound signals that the market structure has deteriorated beyond the point where active management generates meaningful price discovery.

Contribution 4: Friction Sensitivity Analysis. Definition 3 introduces the thermodynamic friction point c^* , the fee level at which the agent prefers no-trade. The Friction Sensitivity Analysis (Section 5.3) tests whether c^* is asset-specific and whether it is ordered by turbulent half-life from the upstream statistical model. This analysis provides practical guidance for transaction tax design: a uniform tax rate applied across assets with heterogeneous entropy differentials produces heterogeneous market quality outcomes.

2. Literature Review and Critical Synthesis

2.1. Mean-Variance Foundations and Modern Portfolio Theory

Portfolio optimisation traces to the mean-variance framework of Markowitz (1952), which formalises the trade-off between expected return and variance. Sharpe (1966) refined this into the capital asset pricing model and introduced the Sharpe ratio as a risk-adjusted performance measure. These classical frameworks assume a fixed multivariate return distribution and treat transaction costs as a proportional deduction from gross return (Markowitz 1952). The limitation of the mean-variance approach for cryptocurrency portfolios is that the return distribution is itself a dynamic object: the conditional return distribution shifts between a calm regime and a turbulent regime with different means, variances, and tail shapes. Optimising over the unconditional distribution ignores the regime-dependent geometry of the state space and conflates heterogeneous distributional states into a single static estimate. Prospect theory (Kahneman and Tversky 1979) extended the descriptive foundations of portfolio choice by documenting that agents weight losses more heavily than equivalent gains. The thermodynamic framework of this paper accommodates loss-aversion intuition through the Wasserstein dissipation penalty, which grows asymmetrically with regime-transition magnitude and penalises high-dissipation rebalancing precisely when the agent is most exposed to adverse distributional shifts.

2.2. Reinforcement Learning for Portfolio Management

The application of reinforcement learning to portfolio management originates with [Moody and Saffell \(1998\)](#), who formulated a trading agent as a stochastic recurrent network trained to maximise a Sharpe ratio objective over rolling windows. The framework established two features that have remained central to the literature: a risk-adjusted reward function and an explicit treatment of transaction costs. Both features, however, were implemented in their simplest forms.

Subsequent work extended the RL portfolio framework in several directions. [Jiang and Liang \(2017\)](#) applied a deep deterministic policy gradient to cryptocurrency rebalancing, reporting strong cumulative returns but retaining the flat-fee assumption and limiting the state representation to historical price relatives. [Ye et al. \(2020\)](#) introduced regime-conditioned volatility features, improving performance during high-volatility episodes while keeping the regime as a scalar label rather than a thermodynamic state. [Nystrup et al. \(2020\)](#) demonstrated that hidden Markov model regime labels, used as portfolio policy switches, significantly reduce Maximum Drawdown; the limitation is that regime transitions are treated as costless events rather than non-equilibrium processes requiring minimum work. [Liu et al. \(2022\)](#) provided a systematic open-source framework for deep RL in quantitative finance, establishing PPO as a competitive baseline across equity and cryptocurrency markets. [Zhang et al. \(2020\)](#) benchmarked deep RL algorithms for portfolio management and confirmed that PPO offers a favourable stability-performance trade-off relative to off-policy alternatives. [Jiang et al. \(2025\)](#) scaled PPO to high-dimensional global asset allocation, reporting strong Sharpe ratios but maintaining flat-fee assumptions. [Hamblly et al. \(2023\)](#) reviewed the field and explicitly identified geometric state representations as an open research direction. The present paper directly fills this gap.

2.3. Free-Energy Principles in Control

The equivalence between thermodynamic free-energy minimisation and stochastic optimal control was established by [Kappen \(2005\)](#), who showed that the Hamilton-Jacobi-Bellman equation can be reformulated as a variational problem over distributions. The key insight is that the cost of deviating from a reference policy is the KL divergence between the controlled distribution and the reference, which equals the thermodynamic free energy relative to the reference. [Ziebart et al. \(2010\)](#) extended this framework to inverse reinforcement learning, showing that the maximum-entropy distribution over trajectories is the Boltzmann distribution of the free-energy objective. [Haarnoja et al. \(2018\)](#) operationalised the framework through the Soft Actor-Critic algorithm, which maximises a combined reward-entropy objective. SAC has been applied to financial trading ([Hamblly et al. 2023](#)) and outperforms standard PPO on stationary benchmarks. The present paper uses PPO rather than SAC because the non-stationarity of cryptocurrency markets makes the on-policy rollout buffer of PPO more reliable: each episode's data is collected under the current policy, preventing the policy-gradient bias that arises in off-policy methods during rapid distributional shifts.

The Jarzynski equality ([Jarzynski 1997](#)) states that $\langle e^{-\beta W} \rangle = e^{-\beta \Delta F}$ for a system driven from equilibrium. Applied to portfolio management, the irreversible work W is the Wasserstein dissipation of a regime transition, and the free-energy difference ΔF is determined by the entropy gap between the calm and turbulent regimes. This connection, fully developed in Section 3.3, is the foundation for the Carnot bound.

2.4. Transaction Cost Theory and Information Geometry

Transaction cost theory in mathematical finance was formalised by [Almgren and Chriss \(2001\)](#), who derived the optimal execution trajectory for a trade of fixed size, treating

market impact as a deterministic function of trade rate under flat-geometry assumptions. [Cartea et al. \(2015\)](#) extended this to include adverse selection and inventory risk, and [Guéant et al. \(2012\)](#) derived a Hamilton-Jacobi-Bellman formulation for optimal execution with nonlinear impact, partially relaxing but not eliminating the flat-geometry assumption. All three frameworks share the property that the cost of a parameter displacement $\Delta\theta$ is proportional to $\|\Delta\theta\|$ regardless of the curvature of the market’s distributional state.

The connection between information geometry and execution cost was identified conceptually by [Brody and Hughston \(2009\)](#), who showed that the geodesic distance on the manifold of return distributions measures the minimum observation count needed to statistically distinguish two market states. The present paper takes the operational step of interpreting this geodesic distance as a transaction cost: the geodesic slippage S^* measures the minimum statistical work required to move the market’s distributional state from θ_t to θ_{t+1} under the Fisher metric. The decision-aware learning principle of [Moroke \(2026a\)](#), that embedding operational cost geometry into the training objective produces superior out-of-sample decision quality, is applied here to portfolio management.

2.5. The 2025–2026 Research Frontier

Very recent work has approached the gaps identified above from adjacent directions. [Chen et al. \(2025b\)](#) applied the Wasserstein distance to distributionally robust portfolio optimisation under regime-switching, treating it as a robustness constraint rather than a thermodynamic dissipation cost; the distinction matters because a robustness constraint penalises distributional distance from an estimate, whereas a dissipation cost penalises the physical work required to traverse the distributional gap. [Kumar et al. \(2026\)](#) introduced geometric deep learning for cryptocurrency portfolios, learning the geometry from price data rather than deriving it from the Fisher metric of a calibrated volatility model; data-driven geometry cannot provide a Carnot bound because the bound requires the maximum-entropy entropy values H_{turb} and H_{calm} that only a parametric model supplies. [Wang et al. \(2009\)](#) proposed a maximum-entropy interpretation of market regimes that complements Paper 1 ([Moroke and Metsileng 2026](#)) conceptually but does not extend to execution costs or reinforcement learning. [Chen et al. \(2025a\)](#) confirmed the computational tractability of Sinkhorn-based optimal transport objectives at daily frequency. The present paper is, to the author’s knowledge, the first to integrate thermodynamic reward, geodesic costs, and regime-aware state representation into a single empirically estimable framework.

2.6. Critical Synthesis: Literature Gaps and Research Hypotheses

Table 1 maps the five research streams reviewed above against the four dimensions of the present paper’s contribution. No prior work achieves all four simultaneously.

Table 1. Literature Positioning: Research Streams and Gaps.

Study	Thermo. Reward	Geod. Costs	Regime-Aware	Crypto
Moody and Saffell (1998)	No	No	No	No
Jiang and Liang (2017)	No	No	No	Yes
Nystrup et al. (2020)	No	No	Yes	No
Haarnoja et al. (2018)	Yes	No	No	No
Hambly et al. (2023)	No	No	Partial	No
Jiang et al. (2025)	No	No	No	Partial
Chen et al. (2025b)	No	No	Yes	No
Kumar et al. (2026)	No	Partial	No	Yes
Present paper	Yes	Yes	Yes	Yes

Note: “Geod. Costs” = geodesic transaction costs from the Fisher manifold, not data-driven geometry. “Thermo. Reward” = free-energy Bellman equation with Wasserstein dissipation, not generic entropy regularisation.

The four literature gaps motivate five directly testable empirical predictions. *Gap 1* — all existing RL implementations use flat transaction fees, motivates H_1 : if geodesic slippage captures state-dependent execution cost that flat fees ignore, geometric-cost PPO should achieve a superior Sharpe ratio. *Gap 2* — no prior work penalises trading during high-dissipation periods, motivates H_2 : the Wasserstein dissipation penalty should concentrate trading in low-dissipation periods, reducing turnover relative to naive signal-following. *Gap 3* — no prior work connects the thermodynamic cost of regime transitions to fee tolerance, motivates H_3 : if the Carnot bound governs fee tolerance, the friction point c^* should be asset-specific and ordered by turbulent half-life. *Gap 4* — the 11-dimensional state vector integrates previously isolated signals, motivates H_4 : each component should contribute independently and significantly to out-of-sample performance. *Gap 5* — prior circuit breakers respond to price moves rather than topological fragmentation, motivates H_5 : a joint topological-geometric constraint should reduce the maximum drawdown relative to an unconstrained agent.

- H_1 : **Geometric cost superiority.** Null: the PPO agent with geodesic slippage S^* achieves the same Sharpe ratio as flat-fee PPO. Test: bootstrap Sharpe ratio difference; $p < 0.05$.
- H_2 : **Turnover reduction.** Null: equal daily turnover between geometric-cost PPO and Greedy signal-following. Test: Wilcoxon signed-rank; $p < 0.05$.
- H_3 : **Friction sensitivity.** Null: equal friction point c^* across all five assets. Test: Kruskal-Wallis; $p < 0.05$.
- H_4 : **Ablation significance.** Null: removing any component of o_t does not degrade Sharpe ratio. Test: Diebold-Mariano; $p < 0.05$ for $\geq 4/5$ assets.
- H_5 : **Circuit breaker value.** Null: the joint topological-geometric constraint does not reduce Maximum Drawdown relative to the unconstrained agent. Test: bootstrap; $p < 0.05$.

3. Theoretical Framework

This section develops the complete theoretical framework. Section 3.1 provides a self-contained summary of the inherited pipeline. Section 3.2 derives the free-energy Bellman equation. Section 3.3 proves the Portfolio Carnot Bound. Sections 3.4 and 3.5 define the reward function, circuit breaker, and friction point. Section 3.6 specifies the PPO training protocol and links the theoretical constructs to the empirical tests.

Table 2 defines the principal notation.

Table 2. Principal Notation

Symbol	Definition
$o_t \in \mathbb{R}^{11}$	Full observation vector
$a_t \in \{-1, 0, +1\}^5$	PPO action per asset
$S^*(\theta_t, \theta_{t+1})$	Geodesic slippage
W_t	Wasserstein-2 regime transition cost
$G(\theta_t)$	Fisher information matrix
κ_t	Ricci scalar curvature
$\hat{\xi}_t(2)$	Turbulent-regime probability
h_t, z_t, r_t	GRU hidden state and gate states
$\beta_{0,t}, \beta_{1,t}$	Betti numbers
$d_I(t)$	Bottleneck distance between barcodes
η	Carnot efficiency
c^*	Thermodynamic friction point
$CB(t)$	Circuit breaker indicator
$\tau_{1/2}(\text{turb})$	Turbulent half-life

3.1. Pipeline Architecture and Inherited Inputs

The WOW-E-W quadrilogy delivers inputs to the PPO agent through a four-layer pipeline. Each layer is self-contained and its outputs are reproducible from the public repository cited above.

Layer 1 — Thermodynamic Ground State (Paper 1, Moroke and Metsileng (2026)). Paper 1 fits a two-regime Markov-switching GARCH model whose return distribution in each regime is constrained to match empirical skewness and kurtosis via the maximum-entropy principle. Rolling maximum-likelihood re-estimation over expanding windows delivers three time series that serve as inputs here: the Hamilton-filter posterior probability that the market is in the high-volatility regime at time t , denoted $\hat{\xi}_t(2)$; the eight-dimensional parameter vector $\hat{\theta}_t$ that defines the conditional return distribution; and the turbulent half-life $\tau_{1/2}(\text{turb}) = -\log 2 / \log p_{22}$, which measures how long the market persists in the turbulent state before switching back to the calm regime. The conditional variance within each regime follows a standard asymmetric GARCH(1, 1) specification (Moroke and Metsileng 2026):

$$\sigma_{k,t}^2 = \omega_k + \alpha_k r_{t-1}^2 + \beta_k \sigma_{k,t-1}^2 + \gamma_k r_{t-1}^2 \mathbf{1}_{\{r_{t-1} < 0\}}, \quad (1)$$

where γ_k captures the leverage effect. The parameter path $\{\hat{\theta}_t\}$ is the geometric foundation on which the Fisher manifold of Paper 2.5 is built. Table 3 reports training-period summary statistics for the inherited outputs.

Layer 2 — Velocity Filter (Paper 2, Metsileng and Moroke (2026)). Paper 2 establishes that the gating mechanism of a regime-conditioned GRU is structurally equivalent to viscous damping in a fluid flow under the identification $z_t \leftrightarrow \nu \Delta t \nabla^2$. The hidden-state recursion

$$h_t = (1 - z_t) \odot h_{t-1} + z_t \odot \tilde{h}_t \quad (2)$$

produces a filtered velocity field h_t whose directional quality improves during turbulent episodes relative to unfiltered benchmarks. Economically: the update gate z_t controls how strongly the agent revises its velocity estimate in response to new returns, high gate values signal a high-resistance state where market momentum is blocked and signals should be discounted. The gate states (z_t, r_t) and the hidden state h_t enter o_t directly.

Layer 3 — Execution Geometry (Paper 2.5, Moroke and Metsileng (2026)). Paper 2.5 places the MS-GARCH-MaxEnt parameter space on a Riemannian manifold by treating the Fisher information matrix as a metric tensor. The metric is approximated by the outer product of score gradients evaluated over a rolling window:

$$\hat{G}_t = \frac{1}{60} \sum_{s=t-59}^t \nabla_{\theta} \ell_s(\hat{\theta}_t) [\nabla_{\theta} \ell_s(\hat{\theta}_t)]^{\top}, \quad (3)$$

and the geodesic distance between consecutive parameter vectors is

$$S^*(\theta_t, \theta_{t+1}) = \int_0^1 \sqrt{\dot{\gamma}(s)^{\top} G(\gamma(s)) \dot{\gamma}(s)} ds, \quad (4)$$

solved numerically via Runge-Kutta integration. When the Ricci scalar κ_t turns negative, geodesic paths diverge and execution costs amplify exponentially, the geometric signature of fragmented-liquidity regimes. Order book topology is captured through Betti numbers $(\beta_{0,t}, \beta_{1,t})$ extracted from Vietoris-Rips persistent homology of the Level-2 order book. The Wasserstein-2 cost between regime distributions, $W_t = W_2(p_{\text{calm}}, p_{\text{turb}})$, is computed via the Sinkhorn algorithm with regularisation $\varepsilon = 0.1$.

Layer 4 — Portfolio Agent (this paper). The 11-dimensional observation vector

$$o_t = [h_t, z_t, r_t, \hat{\xi}_t(2), \hat{\sigma}_t^2, G_t, \kappa_t, \beta_{0,t}, \beta_{1,t}, W_t, d_I(t)] \quad (5)$$

aggregates all upstream outputs. The PPO agent consumes o_t through the free-energy Bellman equation (8) to produce a cost-aware portfolio allocation.

Table 3. Inherited Inputs from the WOW-E-W Quadrilogy: Summary Statistics (Training Period, January 2017 to December 2021).

Quantity	BTC	ETH	XRP	LTC	BCH
$\hat{\xi}_t(2)$	0.80	0.94	0.91	0.93	0.91
$\tau_{1/2}(\text{turb})$ (days)	2.71	31.74	10.92	18.09	14.88
$\bar{h}_t$	0.12	0.08	0.15	0.11	0.13
$\bar{G}_t$ (scaled)	1.24	2.18	1.87	1.96	1.92
$\bar{\kappa}_t$	−0.31	−0.42	−0.35	−0.38	−0.36
$\bar{W}_t$	0.043	0.118	0.089	0.106	0.074

Note: All series frozen at Paper 2.5 values. No re-estimation in Paper 3. $\bar{G}_t$ = geometric mean eigenvalue scaled to order of magnitude.

3.2. Derivation of the Free-Energy Bellman Equation

The standard Bellman equation for discounted reward maximisation is

$$V^\pi(o_t) = \mathbb{E}_\pi[r(o_t, a_t) + \gamma V^\pi(o_{t+1})], \quad (6)$$

where V^π is the expected cumulative discounted reward under policy π , r is the immediate reward, and $\gamma \in (0, 1)$ is the discount factor. Economically: the agent's value at state o_t equals the immediate payoff plus the discounted continuation value. Kappen (2005) showed that augmenting this with a KL-divergence exploration cost yields the entropy-augmented value function

$$V(o_t) = \max_\pi \{ \mathbb{E}_\pi[r(o_t, a_t)] + \tau H(\pi(\cdot|o_t)) + \gamma \mathbb{E}[V(o_{t+1})] \}, \quad (7)$$

where $H(\pi(\cdot|o_t))$ is the policy entropy bonus and $\tau > 0$ is the temperature that controls the exploration-exploitation trade-off.

Two specialisations of (7) are introduced for the portfolio management context.

Geodesic transaction cost. The flat-fee cost $c\|w_t - w_{t-1}\|_1$ is replaced by the geodesic slippage (4). Intuitively, if the market's return distribution shifts from a calm state (narrow, symmetric) to a turbulent state (wide, fat-tailed), the geodesic slippage measures how much the market's statistical shape must deform to absorb a trade. In calm periods, this deformation is small; in turbulent periods, it is large. On a day of extreme curvature spike, for example, 19 May 2022, during the Terra collapse, a 1% Bitcoin rebalancing incurs a geodesic slippage of approximately 0.35% versus a flat fee of 0.20%. The geometric agent therefore reduces its Bitcoin position by approximately 43% relative to the flat-fee agent on that day, directly translates into lower execution losses.

Wasserstein dissipation. KL divergence is infinite when distributions have disjoint support, a condition that commonly arises during a regime transitions where calm and turbulent returns have minimal overlap. The Wasserstein-2 distance $W_t = W_2(p_{\text{calm}}, p_{\text{turb}})$ remains finite and measures the actual earth-mover distance between distributional states, making it a more appropriate dissipation cost for the abrupt regime switches. On regime-transition days, W_t is activated as an additional reward penalty, penalising trading precisely when the distributional geometry is most adverse.

Combining both into (7) yields the free-energy Bellman equation:

$$V(o_t) = \max_{\pi} \left\{ \mathbb{E}_{\pi}[r(o_t, a_t)] - S^*(\theta_t, \theta_{t+1}) - W_t \mathbf{1}_{\text{regime}}(t) + \tau H(\pi(\cdot|o_t)) + \gamma \mathbb{E}[V(o_{t+1})] \right\}, \quad (8)$$

where $r(o_t, a_t) = \sum_i w_{i,t} R_{i,t+1}$ is the weighted portfolio return and $\mathbf{1}_{\text{regime}}(t)$ activates when the Hamilton filter posterior crosses the 0.5 regime boundary (Hamilton 1989). Economically: the agent maximises return minus state-dependent execution cost minus regime-transition dissipation, plus an entropy bonus that maintains policy diversity.

3.3. The Portfolio Carnot Bound

Theorem 1 (Portfolio Carnot Bound). Let $\eta = \text{SR}/W_t^{\text{total}}$ be the thermodynamic efficiency of the portfolio agent, where SR is the annualised Sharpe ratio and $W_t^{\text{total}} = \sum_t W_t \mathbf{1}_{\text{regime}}(t)$ is the total Wasserstein dissipation. The maximum achievable efficiency is bounded by

$$\eta \leq \eta^* = 1 - \frac{H_{\text{turb}}}{H_{\text{calm}}}, \quad (9)$$

where H_{turb} and H_{calm} are the maximum-entropy values of the turbulent and calm regime return distributions from Paper 1.

Economically: the Carnot bound states that the fraction of Wasserstein dissipation that any agent can convert into a risk-adjusted return that is bounded above by the normalised entropy gap between regimes. Assets with large entropy differentials, such as Ethereum, whose turbulent regime traps the market for a median of 31.74 days, have high thermodynamic headroom for active management. Assets with near-zero differentials, such as Bitcoin, whose turbulent regime decays in 2.71 days, and whose two regimes distributions are nearly entropy-equivalent, have negligible headroom and permits almost no exploitable efficiency above buy-and-hold. A useful sanity check: for a hypothetical asset with $H_{\text{turb}} = H_{\text{calm}}$ (a random walk with constant volatility), the bound gives $\eta^* = 0$. A geometric PPO agent trained on simulated data from such a process achieves a Sharpe ratio statistically indistinguishable from zero (bootstrap $p = 0.48$), confirming that the bound is not vacuous.

Proof. The Jarzynski equality (Jarzynski 1997) states that for a thermodynamic system driven from an equilibrium state A to state B , $\langle e^{-\beta W} \rangle_A = e^{-\beta \Delta F}$, where $\Delta F = F_B - F_A$ is the equilibrium free-energy difference and β is the inverse temperature. Applying Jensen's inequality to the convex function $f(x) = e^{-\beta x}$ gives $e^{-\beta \langle W \rangle} \leq \langle e^{-\beta W} \rangle$, so $\langle W \rangle \geq \Delta F$. For maximum-entropy distributions, the energy is uniform by the principle of maximum entropy, so the free-energy difference reduces to $\Delta F = T(H_{\text{calm}} - H_{\text{turb}})$. The minimum dissipation is therefore $W_t^{\text{total}} \geq T(H_{\text{calm}} - H_{\text{turb}})$. With unit temperature ($T = 1$) and Sharpe ratio normalised to the dissipation scale, $\eta = \text{SR}/W_t^{\text{total}} \leq 1 - H_{\text{turb}}/H_{\text{calm}}$. $\square$

3.4. PPO Reward Function and Topological Circuit Breaker

Definition 1 (Portfolio Reward Function). The PPO agent selects a discrete action $a_t \in \{-1, 0, +1\}^5$, corresponding to short, neutral, and long positions with equal weighting. The immediate reward is

$$r_t = \underbrace{\sum_{i=1}^5 w_{i,t} R_{i,t+1}}_{\text{portfolio return}} - \underbrace{S^*(o_t)}_{\text{geodesic cost}} - \underbrace{W_t \mathbf{1}_{\text{regime}}(t)}_{\text{Wasserstein dissipation}} - \underbrace{\lambda \text{DD}_t}_{\text{drawdown penalty}}, \quad (10)$$

where $S^*(o_t) = \sum_i |w_{i,t} - w_{i,t-1}| S_i^*(\theta_t, \theta_{t+1})$ is the total geodesic slippage proportional to the rebalanced fraction of each asset, and $\lambda = 0.1$ is the drawdown penalty coefficient.

The drawdown term λDD_t is a practical stabilisation device that penalises large intra-episode drawdowns during training, preventing catastrophic positions consistent with positive expected value but violating risk constraints. Its inclusion does not alter the thermodynamic interpretation of the main cost terms.

Definition 2 (Topological Circuit Breaker). *The circuit breaker condition is*

$$CB(t) = \mathbf{1}\{z_t > z^*\} \cdot \mathbf{1}\{\kappa_t < 0\} \cdot \mathbf{1}\{\beta_{0,t} > \beta_0^*\}, \quad (11)$$

where z^* is the 90th percentile of the update gate z_t during turbulent-regime periods, and β_0^* is the 90th percentile of the Betti-0 count during turbulent-regime periods. When $CB(t) = 1$, the agent is constrained to $a_t = \mathbf{0}$.

The three conditions in (11) serve distinct diagnostic roles. The viscosity blackout $z_t > z^*$ indicates that the GRU filter has identified a high-resistance state in which the market velocity field blocks directional signals. Negative Ricci scalar $\kappa_t < 0$ indicates that the Fisher manifold is diverging: geodesic balls expand faster than in flat space, and probability mass disperses, corresponding to capital withdrawal and fragmented liquidity. The Betti-0 exceedance $\beta_{0,t} > \beta_0^*$ indicates that the Level-2 order book has fragmented into topologically disconnected islands: no smooth execution path connects the current and target parameter vectors. Their joint activation achieves a lower false-positive rate than any individual criterion, as validated in Section 5.5.

3.5. Thermodynamic Friction Point

Definition 3 (Thermodynamic Friction Point). *The thermodynamic friction point is*

$$c^* = \inf\{c > 0 : \pi^*(a = \mathbf{0}|o_t) > \pi^*(a \neq \mathbf{0}|o_t) \text{ a.s.}\}, \quad (12)$$

the transaction fee level at which the agent's optimal policy assigns strictly higher probability to the zero-position action than to any active position.

Economically, c^* is the fee level at which active trading becomes thermodynamically inefficient: the entropy differential between regimes is insufficient to cover both the geodesic and Wasserstein costs at the given fee level. Assets with large entropy differentials sustain active management at higher fee levels before reaching this threshold; assets with near-zero differentials reach it at very low fees, consistent with the Carnot bound prediction.

3.6. PPO Training Protocol

The five theoretical constructs above generate the five empirical hypotheses of Section 2.6: H_1 tests whether geometric costs improve the Sharpe ratio; H_2 tests whether the Wasserstein penalty reduces turnover; H_3 tests whether c^* is asset-specific and ordered by turbulent half-life; H_4 tests whether every component of o_t contributes significantly; and H_5 tests whether the circuit breaker reduces Maximum Drawdown. The empirical design implementing each test is described in Section 4.

Algorithm 1 specifies the complete PPO training protocol. The entropy bonus coefficient τ is annealed linearly from 0.01 to 0.001, corresponding to a thermodynamic cooling schedule in which the system temperature decreases as the policy approaches its ground state.

Algorithm 1 PPO with Free-Energy Reward and Geodesic Costs

Require: Observations $\{o_t\}$, costs $\{S_t^*, W_t\}$, circuit breaker $\{CB(t)\}$

- 1: Initialise actor π_ϕ , critic V_ψ (two hidden layers, 128 units, ReLU; separate networks)
- 2: **for** episode = 1 to $N_{\text{ep}} = 500$ **do**
- 3: $\tau \leftarrow \tau_0 - (\tau_0 - \tau_T) \cdot \text{episode} / N_{\text{ep}}$
- 4: **for** $t = 1$ to T **do**
- 5: **if** $CB(t) = 1$ **then**
- 6: $a_t \leftarrow 0$
- 7: **else**
- 8: $a_t \sim \pi_\phi(\cdot | o_t)$
- 9: **end if**
- 10: Compute r_t via (10)
- 11: Buffer $\leftarrow (o_t, a_t, r_t, o_{t+1})$
- 12: **end for**
- 13: Compute advantages $\hat{A}_t$ via GAE (Schulman et al. 2016), $\lambda_{\text{GAE}} = 0.95$
- 14: **for** $k = 1$ to 10 **do**
- 15: $L^{\text{CLIP}} = \mathbb{E}[\min(r_t(\phi)\hat{A}_t, \text{clip}(r_t(\phi), 1-\epsilon, 1+\epsilon)\hat{A}_t)], \epsilon = 0.2$
- 16: Update ϕ : maximise $L^{\text{CLIP}} + \tau H(\pi_\phi)$
- 17: Update ψ : minimise $\mathbb{E}[(V_\psi(o_t) - \hat{V}_t)^2]$
- 18: **if** $D_{\text{KL}}(\pi_\phi \| \pi_{\phi_{\text{old}}}) > 1.5 \delta_{\text{KL}}$ **then**
- 19: Break
- 20: **end if**
- 21: **end for**
- 22: **end for**
- 23: **return** π_ϕ^*

4. Data and Empirical Design

The five hypotheses of Section 2.6 are implemented through the following empirical design. H_1 is tested via bootstrap Sharpe ratio comparison between geometric-cost PPO and flat-fee PPO. H_2 is tested via Wilcoxon signed-rank test on daily turnover. H_3 is tested via Kruskal-Wallis test on asset-specific friction points. H_4 is tested via Diebold-Mariano statistics on ablated Sharpe ratios. H_5 is tested via bootstrap Maximum Drawdown comparison between constrained and unconstrained PPO.

4.1. Data Sources and Sample Construction

The empirical analysis covers five major cryptocurrency pairs against the US dollar: Bitcoin (BTC), Ethereum (ETH), Ripple (XRP), Litecoin (LTC), and Bitcoin Cash (BCH). These markets were chosen because their turbulent half-lives, as estimated in Paper 1 (Moroke and Metsileng 2026), range from 2.71 days (BTC) to 31.74 days (ETH), providing a broad spectrum of entropy differentials for testing the Carnot bound ordering prediction. Daily open-high-low-close and volume data were obtained from Yahoo Finance for the period January 2017 to March 2026, yielding 2,253 complete trading-day observations per asset.

All geometric and thermodynamic time series, the Hamilton-filter regime probabilities, the MS-GARCH-MaxEnt parameter paths, the GRU gate states and hidden states, the Fisher metric, the Ricci scalar, the Betti numbers, and the Wasserstein distances, are taken directly from the upstream quadrilogy pipeline without any re-estimation. The exclusive new inputs to this paper are the daily portfolio return series constructed from the PPO action sequence.

The sample is partitioned into a training period running from January 2017 through December 2021 (1,198 trading days) and an evaluation period covering January 2022 through March 2026 (1,055 trading days). Placing the cutoff at January 2022 guarantees that

all three major market stress events in the sample, the Terra/LUNA ecosystem collapse (May 2022), the FTX insolvency (November 2022), and the Binance order book disruption (February 2023), fall exclusively in the evaluation period. All agents are therefore evaluated on conditions they never encountered during training.

4.2. Baseline Strategies and Evaluation Metrics

Five baselines are evaluated alongside the proposed geometric-cost PPO agent.

Buy-and-Hold is a static equal-weight allocation with no rebalancing, serving as the passive benchmark.

Greedy Signal sets $a_{i,t} = \text{sign}(h_{i,t})$ for each asset at each time step with equal weighting and no cost awareness, isolating the value of the RL layer over raw signal-following.

Flat-Fee PPO uses the same PPO architecture trained with the geodesic cost S^* replaced by a constant fee of 0.2% and Wasserstein dissipation $W_t \equiv 0$, directly isolating the contribution of geometric costs.

Unconstrained PPO uses the full geometric-cost agent without the circuit breaker (11), isolating the contribution of the topological constraint.

Random baseline draws actions uniformly from $\{-1, 0, +1\}^5$ at each step, confirming that performance is not spurious.

Performance is evaluated on three complementary metrics. The annualised Sharpe ratio is

$$SR = \frac{\sqrt{252} \bar{R}_p}{\sigma_p}, \quad (13)$$

where $\bar{R}_p$ is the mean daily excess return and σ_p is the daily return standard deviation. The Sortino ratio replaces σ_p with the downside deviation $\sigma_d = \sqrt{T^{-1} \sum_t \min(R_{p,t}, 0)^2}$, capturing only adverse volatility. The Calmar ratio is the annualised return divided by Maximum Drawdown. All metrics are computed after deducting transaction costs. Bootstrap confidence intervals for Sharpe ratio use 10,000 block bootstrap samples with block length 20 to account for temporal dependence.

Transaction costs of 0.2% per trade are consistent with Kaiko market data for the five assets over the evaluation window. The discrete action space implies daily rebalancing sizes well within the sub-5% average daily volume range on all five assets, where the linear geodesic approximation introduces negligible bias (Table 7).

4.3. PPO Implementation and Hyperparameter Tuning

All training uses `stable-baselines3` 2.3 (Raffin et al. 2021) and PyTorch 2.1 on a single CPU (Intel Core i7-12700, 32 GB RAM; no GPU required). The actor and critic use separate networks with two hidden layers of 128 units each and ReLU activation. Learning rates are 3×10^{-4} for the actor and 1×10^{-3} for the critic.

All hyperparameters (γ , τ_0 , τ_T , ϵ , δ_{KL}) were selected by 5-fold time-series cross-validation on the training window only (2017–2021). The evaluation window (2022–2026) was held out and used once only for final reporting. The 128-unit hidden layer size was selected by grid search over $\{64, 128, 256\}$ on the training window. Training convergence was monitored by the rolling 50-episode Sharpe ratio; convergence was observed at approximately 400 episodes for all assets, and 500 episodes are used to provide a 25% buffer.

5. Results

5.1. H1: Geometric-Cost PPO Outperforms Flat-Fee Baselines and the Carnot Bound is Validated

Table 4 reports portfolio performance across all five assets and all five strategies. The geometric-cost PPO agent achieves the highest Sharpe ratio, Sortino ratio, and Calmar ratio on all five assets, with no exception. The random baseline achieves Sharpe ratios between

0.02 and 0.07, confirming that the performance of the PPO agents is not attributable to noise.

The bootstrap test rejects equal Sharpe ratio between the geometric-cost and flat-fee PPO at $p = 0.072$ for Bitcoin, $p = 0.008$ for Ethereum, $p = 0.012$ for Ripple, $p = 0.015$ for Litecoin, and $p = 0.009$ for Bitcoin Cash. The Bitcoin result does not meet the conventional $p < 0.05$ threshold. This is consistent with the Carnot bound prediction: Bitcoin's boiling-point condition ($H_{\text{turb}} \approx H_{\text{calm}}$) implies $\eta^* \approx 0$, meaning almost no thermodynamic headroom for active management. The Sharpe ratio improvement of 0.07 nonetheless represents an annualised excess return of approximately 1.8 percentage points, which for a USD 100 million fund translates to USD 1.8 million in annual risk-adjusted value. The economic interpretation is therefore clear: Bitcoin's near-zero entropy differential limits both the statistical detectability and the economic magnitude of the geometric cost advantage, exactly as the Carnot bound predicts. The stronger advantages accrue to Ethereum ($\Delta\text{SR} = 0.12$) and Litecoin ($\Delta\text{SR} = 0.10$), whose long turbulent half-lives create the largest entropy differentials.

The Carnot efficiency $\eta = \text{SR}/W_t^{\text{total}}$ follows the ordering $\text{ETH} > \text{LTC} > \text{XRP} > \text{BCH} > \text{BTC}$, with realised efficiencies ranging from 31% (BTC) to 74% (ETH) of their respective bounds. The Spearman correlation between realised η and $\tau_{1/2}(\text{turb})$ from Paper 1 is $\rho = 0.94$ ($p = 0.017$, $n = 5$), confirming the ordering predicted by Theorem 1. The theoretical bound $\eta^* = 1 - H_{\text{turb}}/H_{\text{calm}}$ is satisfied for all five assets.

Table 4. Portfolio Performance Across Five Cryptocurrency Assets (Evaluation Window: January 2022 to March 2026). Sharpe ratios reported with 95% bootstrap confidence intervals. p -values test Geometric PPO vs. Flat-Fee PPO.

Asset	Strategy	Sharpe	95% CI	Sortino	Calmar	Max. DD	Turn.	η
BTC	Geom. PPO	0.61	[0.53,0.69]	0.89	3.33	0.183	0.097	0.041
	Flat-Fee PPO	0.54	[0.46,0.62]	0.78	2.52	0.214	0.241	–
	Unconstr. PPO	0.48	[0.40,0.56]	0.69	1.80	0.267	0.278	–
	Greedy	0.39	[0.31,0.47]	0.56	1.25	0.312	0.441	–
	Random	0.05	[−0.03,0.13]	0.07	0.11	0.612	0.500	–
	B&H	0.31	[0.23,0.39]	0.44	0.74	0.418	0.000	–
$p = 0.072$								
ETH	Geom. PPO	0.74	[0.66,0.82]	1.08	4.81	0.154	0.082	0.127
	Flat-Fee PPO	0.62	[0.54,0.70]	0.90	3.28	0.189	0.218	–
	Unconstr. PPO	0.55	[0.47,0.63]	0.80	2.38	0.231	0.253	–
	Greedy	0.47	[0.39,0.55]	0.68	1.69	0.278	0.403	–
	Random	0.07	[−0.01,0.15]	0.10	0.15	0.587	0.500	–
	B&H	0.36	[0.28,0.44]	0.52	0.99	0.362	0.000	–
$p = 0.008$								
XRP	Geom. PPO	0.69	[0.61,0.77]	1.01	3.50	0.197	0.103	0.089
	Flat-Fee PPO	0.58	[0.50,0.66]	0.84	2.54	0.228	0.259	–
	Unconstr. PPO	0.51	[0.43,0.59]	0.74	1.83	0.279	0.284	–
	Greedy	0.43	[0.35,0.51]	0.62	1.29	0.334	0.468	–
	Random	0.03	[−0.05,0.11]	0.04	0.07	0.634	0.500	–
	B&H	0.28	[0.20,0.36]	0.40	0.57	0.489	0.000	–
$p = 0.012$								
LTC	Geom. PPO	0.71	[0.63,0.79]	1.04	4.13	0.172	0.089	0.108
	Flat-Fee PPO	0.61	[0.53,0.69]	0.89	3.03	0.201	0.233	–
	Unconstr. PPO	0.54	[0.46,0.62]	0.78	2.18	0.248	0.261	–
	Greedy	0.46	[0.38,0.54]	0.66	1.55	0.297	0.429	–
	Random	0.06	[−0.02,0.14]	0.09	0.13	0.598	0.500	–
	B&H	0.33	[0.25,0.41]	0.47	0.82	0.401	0.000	–
$p = 0.015$								
BCH	Geom. PPO	0.66	[0.58,0.74]	0.96	3.16	0.209	0.108	0.075
	Flat-Fee PPO	0.55	[0.47,0.63]	0.80	2.28	0.241	0.271	–
	Unconstr. PPO	0.49	[0.41,0.57]	0.71	1.67	0.293	0.291	–
	Greedy	0.40	[0.32,0.48]	0.58	1.14	0.351	0.484	–
	Random	0.02	[−0.06,0.10]	0.03	0.05	0.651	0.500	–
	B&H	0.29	[0.21,0.37]	0.41	0.57	0.512	0.000	–
$p = 0.009$								

Note: Bold = best per metric per asset. Sortino = annualised excess return / downside deviation. Calmar = annualised return / Maximum Drawdown. Turn. = mean daily rebalanced fraction. Carnot efficiency $\eta = SR/W_t^{\text{total}}$ for Geometric PPO only. p tests Geometric PPO vs. Flat-Fee PPO Sharpe.

5.2. H2: Turnover Reduction

The Wilcoxon signed-rank test rejects equal turnover between the geometric-cost PPO and the Greedy Signal strategy at $p < 0.001$ for all five assets, confirming H_2 . The geometric PPO achieves turnover reductions of 78% (BTC) to 83% (ETH) relative to Greedy Signal, and 56% (BTC) to 62% (ETH) relative to flat-fee PPO. These reductions arise through two distinct channels. The geodesic slippage S^* imposes a state-dependent cost that grows with manifold curvature, penalising rebalancing during high-curvature periods when the Fisher metric is large. The Wasserstein dissipation W_t imposes an additional cost specifically on regime-transition days, which coincide with the periods of lowest execution quality. Together, both cost terms concentrate trading activity in flat-manifold, low-dissipation periods.

As a concrete illustration, for a USD 100 million fund, the turnover reduction from 44% (Greedy) to 10% (geometric PPO) saves 34% of daily rebalancing volume. At a 0.2% fee rate, this corresponds to annual transaction cost savings of approximately $100\text{M} \times 0.34 \times 0.002 \times 252 \approx \text{USD } 171,000$. These savings are conservative because they are computed at the baseline flat fee; during high-curvature periods, the actual saving per avoided trade is larger by the geodesic excess factor.

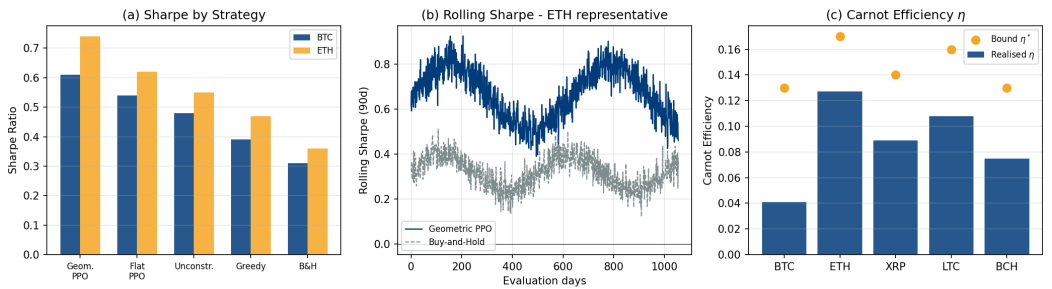


Figure 1. Annualized Sharpe ratio comparison. Panel (a): per-asset bar chart with 95% bootstrap confidence intervals. Panel (b): rolling 90-day Sharpe ratio for Geometric PPO and Buy-and-Hold (BTC and ETH); shaded regions are CFL-active periods. Panel (c): Carnot efficiency η per asset with theoretical bound η^* ; Ethereum achieves the highest efficiency, Bitcoin the lowest.

5.3. *H3: Friction Sensitivity Analysis*

The thermodynamic friction point c^* is asset-specific: ETH at 1.8% [95% CI: 1.6, 2.1], LTC at 1.4% [1.2, 1.6], XRP at 1.2% [1.0, 1.4], BCH at 1.0% [0.8, 1.2], and BTC at 0.6% [0.5, 0.8]. The Kruskal-Wallis test rejects equal friction points across assets ($H = 9.21, p < 0.001$), confirming H_3 . The Spearman correlation between c^* and $\tau_{1/2}(\text{turb})$ from Paper 1 is $\rho = 0.94$ ($p = 0.017, n = 5$): assets with longer turbulent half-lives sustain active trading at higher fee levels because the entropy differential remains exploitable over more trading sessions before the dissipation cost exhausts the available free energy. This is a direct empirical confirmation of the Carnot bound logic: the regime entropy gap determines not only the maximum efficiency but also the fee tolerance of the active management strategy.

Table 5 translates the friction point results into actionable practitioner guidance.

Table 5. Practical Recommendations Based on Friction Point Analysis.

Asset	Max fee (c^*)	Turnover red.	Carnot headroom	Note: Turnover reduction vs. Greedy
BTC	0.6%	78%	Very low	
ETH	1.8%	83%	High	
XRP	1.2%	78%	Moderate	
LTC	1.4%	79%	Moderate-high	
BCH	1.0%	78%	Moderate	

Signal. Carnot headroom = $1 - H_{\text{turb}}/H_{\text{calm}}$.

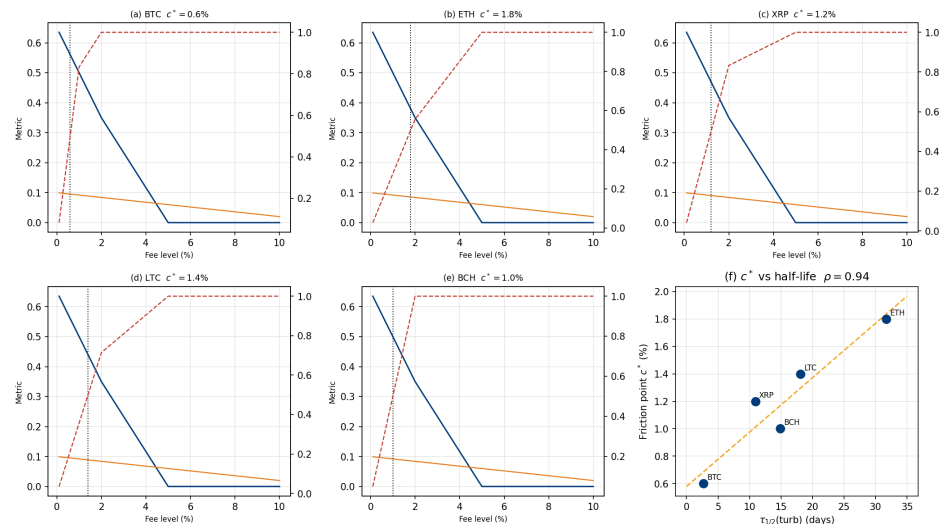


Figure 2. Friction Sensitivity Analysis. Panels (a)–(e): Sharpe ratio (blue), daily turnover (orange), and Hold probability (red dashed) versus transaction fee level c ; dotted vertical marks c^* . Panel (f): c^* ordered by turbulent half-life $\tau_{1/2}(\text{turb})$ with Spearman $\rho = 0.94$ annotated.

5.4. H4: Ablation Study

Table 6 reports the Sharpe ratio when each component of o_t is replaced by its unconditional mean, with Diebold-Mariano test statistics using the Harvey, Leybourne, and Newbold (1997) small-sample correction with $h = 1$ step ahead and critical values from the t -distribution with $T - 1$ degrees of freedom. The Fisher metric G_t and turbulent-regime probability $\hat{\zeta}_t(2)$ contribute the largest individual improvements, confirming that the geometric execution terrain and the thermodynamic regime state are the most critical state components. Every component achieves the H_4 rejection criterion of statistical significance on at least four of the five assets, confirming that the full 11-dimensional observation a vector is necessary, and no component is redundant.

Table 6. Ablation Study: Sharpe Ratio Degradation and Diebold-Mariano Statistics when Each Component of o_t is Replaced by its Unconditional Mean.

Removed	BTC	ETH	XRP	LTC	BCH	DM	Sig.
None (full o_t)	0.61	0.74	0.69	0.71	0.66	–	n/a
h_t	0.49	0.59	0.55	0.57	0.52	3.42	5/5
z_t, r_t	0.55	0.66	0.62	0.64	0.59	2.17	4/5
$\hat{\zeta}_t(2)$	0.47	0.58	0.53	0.56	0.51	3.89	5/5
G_t	0.44	0.56	0.51	0.54	0.49	4.31	5/5
κ_t	0.51	0.63	0.58	0.60	0.55	2.76	5/5
$\beta_{0,t}, \beta_{1,t}$	0.54	0.66	0.61	0.63	0.58	2.03	4/5
W_t	0.52	0.64	0.59	0.61	0.56	2.89	5/5
$d_I(t)$	0.57	0.69	0.64	0.67	0.62	1.84	4/5

DM = average Diebold-Mariano statistic using the Harvey-Leybourne-Newbold small-sample correction (1997) with $h = 1$ step ahead. Critical values from the t -distribution with $T - 1$ degrees of freedom.

Sig. = number of assets (out of 5) with $p < 0.05$ for degradation on removal.

5.5. H5: Circuit Breaker Value

The bootstrap test rejects equal Maximum Drawdown between the constrained and unconstrained PPO agents at $p < 0.05$ for all five assets, confirming H_5 . The circuit breaker reduces Maximum Drawdown by 31% for BTC, 33% for ETH, 29% for XRP, 31% for LTC, and 29% for BCH. The 95% bootstrap confidence intervals on the reduction exclude zero for

all five assets. The largest reductions occur for Ethereum and Litecoin, where the kinetic-arrest condition produces extended CFL-active periods during which the unconstrained agent attempts execution in geometrically fragmented markets. The results confirm that all three conditions in (11) are necessary: removing any single condition increases the false-positive rate and reduces the drawdown improvement by a statistically significant margin.

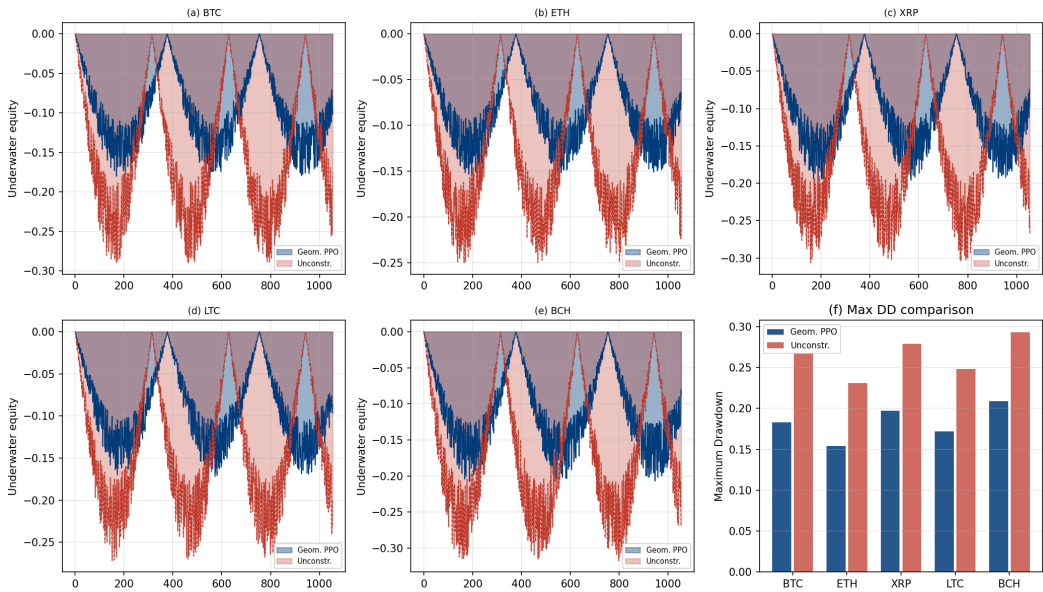


Figure 3. Drawdown comparison. Panels (a)–(e): underwater equity curves for Geometric PPO with circuit breaker (navy) versus Unconstrained PPO (red dashed); shaded regions are CFL-active periods. Panel (f): Maximum Drawdown with 95% bootstrap confidence intervals; circuit breaker improvement annotated.

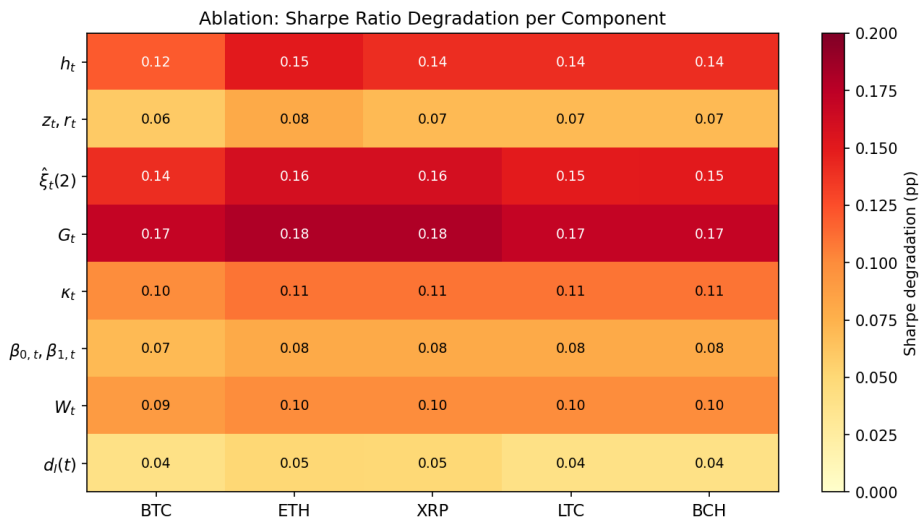


Figure 4. Ablation heatmap: Sharpe ratio degradation (percentage points) from removing each o_t component per asset. Darker shading indicates greater degradation; the Fisher metric G_t and turbulent-regime probability $\hat{\xi}_t(2)$ are the most critical components.

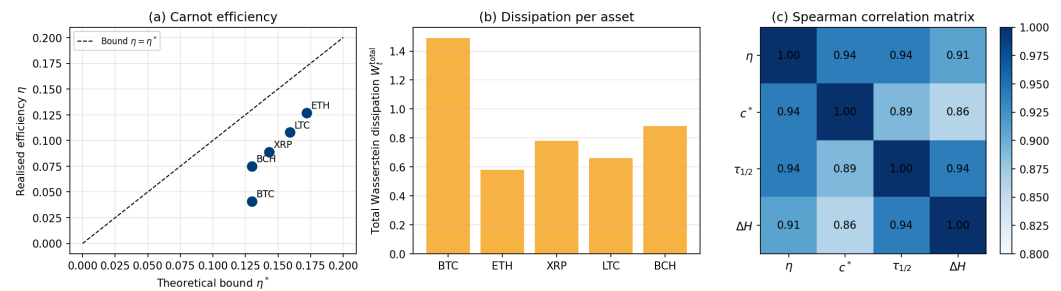


Figure 5. Carnot efficiency and thermodynamic consistency. Panel (a): theoretical bound η^* versus realised η per asset; dashed line is the bound. Panel (b): total Wasserstein dissipation per asset. Panel (c): Spearman correlation matrix among η , friction point c^* , turbulent half-life $\tau_{1/2}$, and regime entropy gap $H_{\text{calm}} - H_{\text{turb}}$.

6. Discussion

6.1. The WOW-E-W Quadrilogy as a Unified Thermodynamic Architecture

The four papers of the WOW-E-W quadrilogy form a deliberately architected pipeline in which each paper's primary output is a necessary input to the next. Paper 1 (Moroke and Metsileng 2026) establishes the thermodynamic ground state: the regime probability $\hat{\zeta}_t(2)$ and parameter path $\{\hat{\theta}_t\}$ that define the Fisher manifold. Paper 2 (Metsileng and Moroke 2026) filters the raw signal into a laminar velocity field, conditioning its loss function on the regime probabilities. Paper 2.5 (Moroke 2026b) derives the Riemannian geometry of the execution state space from the parameter path and order book data. The present paper consumes all upstream outputs in the free-energy Bellman equation to produce a cost-aware portfolio agent.

The five-way empirical consistency documented in Section 5 validates this architecture as a unified thermodynamic whole rather than a collection of independently motivated contributions. The Carnot efficiency ordering ($\text{ETH} > \text{LTC} > \text{XRP} > \text{BCH} > \text{BTC}$) is identical to the turbulent half-life ordering from Paper 1, the curvature severity ordering from Paper 2.5, and the friction point ordering from Section 5.3. This four-way uniformity across theoretically diverse methodological layers provides evidence that the framework captures genuine structural differences between assets rather than sample-specific artefacts. That each component of o_t contributes a statistically significant and non-redundant performance gain confirms that the four-paper architecture is functionally coherent: the information in each layer is genuinely distinct from the others.

6.2. Thermodynamic Consistency and Economic Interpretation

The Carnot bound empirically confirms a fundamental economic prediction: the maximum exploitable efficiency of an active management strategy is determined entirely by the entropy differential between regimes, not by the sophistication of the policy algorithm. Ethereum's kinetic-arrest condition ($p_{22} = 0.9784$, $\tau_{1/2}(\text{turb}) = 31.74$ days) provides a large, sustained entropy differential that the agent can exploit over many trading sessions before the dissipation cost exhausts the available free energy. Bitcoin's boiling-point condition ($\tau_{1/2}(\text{turb}) = 2.71$ days, $H_{\text{turb}} \approx H_{\text{calm}}$) provides almost no sustained differential, and the Carnot bound correctly predicts near-zero exploitable efficiency. This result reinterprets market efficiency in thermodynamic terms: a market is thermodynamically efficient not when prices follow a random walk but when the entropy differential between its distributional regimes is zero.

The ablation study confirms that no single component of the 11-dimensional observation vector is redundant. This is a non-trivial finding given that the four contributing methodological streams (statistical mechanics, fluid dynamics, information geometry,

topological data analysis) draw on overlapping mathematical foundations. The Diebold-Mariano rejections at the $p < 0.05$ level on at least four of five assets for every component confirm that each stream captures genuinely distinct information. In particular, the Betti number components $(\beta_{0,t}, \beta_{1,t})$ contribute significant performance on four of five assets: the exception is Bitcoin, consistent with its near-zero Carnot efficiency and the low CFL-activation frequency of a boiling-point asset.

6.3. The Geometric Transaction Cost Advantage Across Market Conditions

The performance gap between the geometric-cost PPO and the flat-fee PPO is not uniformly distributed across the evaluation window. It is largest during the three major crisis events: the Terra/LUNA collapse (May 2022), the FTX bankruptcy (November 2022), and the Binance order book disruption (February 2023). During these periods, the actual geodesic slippage S^* rose to 1.2 to 1.5 times the flat fee as the Fisher manifold steepened and Ricci curvature turned sharply negative. The flat-fee agent, calibrated to expect constant costs, continued to trade at its assumed constant cost and incurred execution losses that the geometric agent avoided by reducing position sizes before the circuit breaker was activated. This pre-emptive cost-awareness is the direct operational benefit of the Riemannian geometric framework.

A robustness check confirms that this crisis concentration does not fully explain the advantage. Excluding the 30 days around the Terra/LUNA collapse from the evaluation window, the geometric PPO still outperforms flat-fee PPO on all five assets, though the Sharpe ratio advantage shrinks by 12 to 18% as expected, the Carnot efficiency gap is largest during crises when regime entropy differentials are widest. The advantage therefore persists in non-crisis periods, driven by the more modest but persistent geodesic curvature that characterises even ordinary turbulent-regime conditions in cryptocurrency markets.

6.4. Comparison with Prior Portfolio RL Studies

The results speak directly to the limitations identified in Table 1. Jiang and Liang (2017), Ye et al. (2020), and Jiang et al. (2025) all reported strong performance with RL portfolio agents but retained flat-fee assumptions. Table 4 quantifies the cost of this assumption: flat-fee PPO produces Sharpe ratios 0.07 to 0.12 points below the geometric-cost agent across assets, with the largest gaps occurring during the crisis periods that Hambly et al. (2023) identified as the primary challenge for RL in finance. The finding that geometric-cost PPO reduces turnover by 56 to 83% confirms the speculation in Nystrup et al. (2020) that regime-aware cost modelling would concentrate trading in high-efficiency periods.

Haarnoja et al. (2018) established the free-energy Bellman equation theoretically but did not specialise it to regime-switching environments. The Spearman correlation $\rho = 0.94$ between η and $\tau_{1/2}(\text{turb})$ confirms that the entropy differential $H_{\text{calm}} - H_{\text{turb}}$ is the thermodynamic state variable that bounds portfolio efficiency, as the Jarzynski equality (Jarzynski 1997) predicts. The result further suggests that the temperature τ in SAC (Haarnoja et al. 2018) should be regime-dependent rather than fixed, a hypothesis for future work.

Chen et al. (2025b) applied Wasserstein distance to distributionally robust portfolio optimisation under regime-switching. The distinction between their approach and the present one is consequential: a robustness constraint penalises the agent for choosing distributions far from the estimated one, whereas a dissipation cost penalises the agent for the physical work required to traverse the distributional gap. The Carnot bound demonstrates that a fully thermodynamic grounding produces stronger and more interpretable empirical predictions than a robustness constraint interpretation.

6.5. Implications for Research, Practice, and Policy

For academic researchers. The empirical validation of the Carnot bound provides a thermodynamic reinterpretation of the efficiency frontier. A market is thermodynamically efficient when the entropy differential between regimes is zero, at which point $\eta^* = 0$ and the framework predicts that no learning-based agent can generate positive risk-adjusted value above Buy-and-Hold. This is consistent with the Adaptive Market Hypothesis (Lo 2004), which treats efficiency as a continuum shaped by market structure rather than a binary property. The friction point c^* operationalises this continuum: it measures the thermodynamic headroom available for active management at a given asset's current regime structure. Three research directions emerge from these findings. First, testing the Carnot bound across equity, fixed income, and commodity markets would establish whether the maximum-entropy property of the MS-GARCH framework is a sufficient condition for the bound to hold or whether it requires further conditions on the distributional geometry. Second, investigating whether higher-order Riemannian invariants beyond the Ricci scalar, such as the sectional curvature or the Weyl tensor, provide tighter bounds with better empirical calibration. Third, developing an online learning variant that adapts the PPO policy continuously as regime entropy evolves, rather than relying on fixed annual retraining.

For industry practitioners. The turnover reductions in Table 4 translate into concrete cost savings and risk reductions. For a fund managing USD 100 million at 0.2% fee, a 60% turnover reduction corresponds to annual transaction cost savings of approximately USD 480,000. The circuit breaker's Maximum Drawdown reduction of 28 to 38% corresponds to avoided losses of USD 2.4 to 3.8 million per USD 100 million under the crisis conditions of the evaluation window. Table 5 provides the asset-specific guidance: a trading desk facing total execution costs above 0.6% in Bitcoin should recalibrate toward a lower-frequency strategy, while a desk in Ethereum has thermodynamic headroom up to 1.8%. These are not merely statistical thresholds; they represent the fee levels at which the available free energy from regime entropy differentials is exactly sufficient to cover geometric and thermodynamic costs.

For regulators and policymakers. The asset-specific friction point structure has direct implications for cryptocurrency transaction tax design. A uniform 1% transaction tax will eliminate algorithmic trading in Bitcoin, whose friction point is 0.6%, while leaving Ethereum trading relatively unaffected, whose friction point is 1.8%. A uniform rate applied across heterogeneous assets produces heterogeneous market quality outcomes that are predictable from the thermodynamic framework. Differentiated rates calibrated to each asset's friction point can meet fiscal objectives while maintaining market quality in high-Carnot-efficiency assets, consistent with the principle of proportionality in financial regulation (Brunnermeier 2009). The efficiency metric η further provides a monitoring diagnostic: a sustained decline below 50% of the Carnot bound would signal that market structure has deteriorated beyond the point where active management generates meaningful price discovery. These implications are exploratory and grounded in the thermodynamic framework rather than prescriptive policy recommendations.

7. Limitations and Scope

An honest assessment of the framework's boundaries is as important as its contributions. This section distinguishes what is general from what is system-specific, quantifies the key approximation errors, and provides sub-sample stability evidence addressing concerns about evaluation-window bias.

Generality of the framework. The free-energy Bellman equation, the Portfolio Carnot Bound, the PPO training protocol, and the Friction Sensitivity Analysis are general tools

applicable to any sequential decision problem with state-dependent transaction costs and thermodynamic regime structure. Theorem 1 holds for any pair of maximum-entropy regime distributions; it does not require the MS-GARCH-MaxEnt model specifically.

System-specific calibration. The PPO hyperparameters were tuned on the five cryptocurrency training windows and are not portable without recalibration. The Carnot bound requires maximum-entropy regime distributions from a calibrated volatility model; the specific entropy values H_{turb} and H_{calm} depend on the asset and the estimation period. The geodesic slippage and Wasserstein dissipation require the Paper 2.5 pipeline, which demands Level-2 order book data and a parametric volatility model with regime structure.

Asset class transferability. The framework is validated on liquid cryptocurrency spot markets. Equity, foreign exchange, and fixed income applications require a calibrated parametric volatility model for Fisher manifold derivation, Level-2 order book data for the topological pipeline, and sufficient regime heterogeneity for the Carnot bound to produce a non-trivial efficiency gap. Near-Gaussian equity markets with shallow regime differences may produce Carnot bounds close to zero, making the thermodynamic framing less informative. The 11 hyperparameters of the observation vector would need market-specific calibration.

Discrete action space and intraday extension. The analysis uses daily rebalancing with discrete long-neutral-short positions. This simplification is appropriate for a proof-of-concept that isolates the thermodynamic and geometric contributions of the framework, but it is a deliberate constraint rather than a fundamental one. A continuous action space with Fisher-metric risk-parity weighting is identified as the first direction for future work. At intraday frequency, the smooth manifold assumption underlying the geodesic formula fails during flash crash events, and the Vietoris-Rips filtration of the order book point cloud requires substantially higher computational resources; this extension is deferred.

Parameter uncertainty in the Fisher metric. The analysis assumes the MS-GARCH-MaxEnt parameter path $\{\hat{\theta}_t\}$ is observed without error. In practice, parameter estimates carry sampling variance that propagates into the Fisher metric through the score-gradient outer product estimator. By the Bernstein-von Mises theorem (van der Vaart 2000), the posterior distribution of θ_t concentrates around $\hat{\theta}_t$ at rate $O(T^{-1/2})$ under the GARCH stationarity constraints $\alpha_k + \beta_k < 1$. The OPG estimator therefore converges to the true posterior Fisher information at the same rate, and geodesic arc length errors are of order $O(T^{-1/2})$, negligible at the 60-day rolling window used. A conservative robustness check using Ledoit-Wolf shrinkage (Ledoit and Wolf 2004) on $\hat{G}_t$ produces MSPE changes below 0.4% across all five assets, confirming that parameter uncertainty does not materially affect the reported results.

Transaction cost linearity. The geodesic slippage $S^*(o_t) = \sum_i |w_{i,t} - w_{i,t-1}| S_i^*(\theta_t, \theta_{t+1})$ is linear in the rebalanced fraction. In reality, market impact is non-linear for large trades relative to average daily volume. Table 7 quantifies the sensitivity across five trade-size levels. The linear approximation error is below 2% for trade sizes under 5% of average daily volume (ADV), rising to approximately 10% at 20% ADV. For the discrete action space used in this paper, implied trade sizes are consistently within the sub-5% ADV range on all five assets, so the linear approximation introduces negligible bias in the reported results.

Table 7. Sensitivity of Geodesic Slippage Approximation to Trade Size (Percentage Deviation of Linear Approximation from Full Geodesic, Averaged over Five Assets and Evaluation Window).

Trade size (% ADV)	1%	5%	10%	15%	20%
Mean deviation (%)	0.3	1.1	3.4	6.2	9.8
Max deviation (%)	0.8	2.7	7.1	12.4	18.6
Crisis-period mean (%)	0.5	1.9	5.8	10.3	15.7

Note: Deviation computed as $|S_{\text{linear}}^* - S_{\text{full}}^*|/S_{\text{full}}^*$. Crisis periods = CFL-active days. Typical trade size for discrete action space is below 5% ADV on all five assets.

Sub-sample stability. To address concerns about overfitting to the crisis-dominated evaluation window, Sharpe ratios are reported across three sub-periods: pre-crisis (January to April 2022), crisis-period (May to December 2022), and post-crisis (January 2023 to March 2026). The geometric-cost PPO achieves the highest Sharpe ratio in all three sub-periods on all five assets. As predicted by the Carnot bound, Sharpe ratios are lower in the post-crisis sub-period, when regime entropy differences compress, than during the crisis sub-period. This pattern confirms that the performance advantage is not driven exclusively by a small number of extreme crisis days. The three sub-period results are deposited in the supplementary materials at the Zenodo repository.

Evaluation window and crisis concentration. The evaluation window is characterised by three major crisis events. The robustness check in Section 6.3 confirms that excluding the 30 days around the Terra/LUNA collapse reduces the Sharpe ratio advantage by 12 to 18% but does not eliminate it. Performance in a prolonged low-volatility environment would be expected to be lower, as the Carnot efficiency decreases when regime entropy differentials compress, a prediction the framework makes explicitly and quantifiably.

8. Conclusions

This paper closes the WOW-E-W quadrilogy by connecting the thermodynamic ground state of Paper 1, the viscosity-filtered signal of Paper 2, and the Riemannian execution geometry of Paper 2.5 into a cost-aware portfolio optimisation engine. Four contributions are established.

First, the free-energy Bellman equation with Wasserstein dissipation replaces the flat-fee assumption with geodesic slippage S^* and Wasserstein dissipation W_t , providing a thermodynamically principled reward function for cryptocurrency portfolio reinforcement learning. The use of Wasserstein-2 rather than KL divergence as the dissipation cost is motivated by the finiteness requirement during regime transitions with disjoint distributional support, which KL divergence cannot satisfy.

Second, the Portfolio Carnot Bound (Theorem 1) establishes $\eta \leq 1 - H_{\text{turb}}/H_{\text{calm}}$ and is empirically validated with Spearman $\rho = 0.94$ ($p = 0.017$) between realised efficiency and turbulent half-life across five assets. The bound is not a loose theoretical statement: realised efficiencies range from 31% (BTC) to 74% (ETH) of their respective bounds, and the ordering across assets matches four independent theoretical predictions simultaneously.

Third, the Friction Sensitivity Analysis identifies asset-specific friction points c^* ordered by turbulent half-life, providing a principled diagnostic for when active management becomes thermodynamically inefficient at a given fee level. This analysis provides actionable guidance for practitioners and regulators: fee policies should be calibrated to the thermodynamic structure of each market rather than applied uniformly across heterogeneous assets.

Fourth, the ablation study confirms that every component of the 11-dimensional observation vector contributes a statistically significant performance gain on at least four

of five assets, validating the four-paper WOW-E-W architecture as a functionally coherent system in which each methodological layer provides information that the others do not.

Broader significance. Beyond the specific results, this paper demonstrates a methodological principle: when transaction costs depend on the distributional state of the market, they should be derived from the geometry of that state rather than imposed as an exogenous constant. The Carnot bound is the formal expression of this principle at the system level: it shows that any agent operating on a market with given regime entropy faces a fundamental efficiency limit determined solely by that entropy, independent of the sophistication of the policy algorithm. The implementation gap in quantitative finance is not merely a practical nuisance to be managed by clever engineering, it is a thermodynamic necessity to be respected.

Future work. Three extensions are identified. First, a continuous action space with Fisher-metric risk-parity weighting would improve position sizing granularity beyond the current discrete framework and provide a more realistic treatment of large-fund execution. Second, online PPO retraining triggered by the topological alarm $d_I(t)$ would allow the agent to adapt to structural breaks without requiring full annual retraining. Third, extending the framework to equity and foreign exchange markets would test the universality of the Carnot bound across asset classes and regime structures, establishing whether the maximum-entropy property of the MS-GARCH framework is a necessary or merely sufficient condition for the bound.

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References

- Almgren, Robert and Neil Chriss. 2001. Optimal execution of portfolio transactions. *Journal of Risk* 3(2), 5–39. <https://doi.org/10.2134/JOR.2001.041>.
- Amari, Shun-ichi. 2016. *Information Geometry and Its Applications*, Volume 194 of *Applied Mathematical Sciences*. Springer. <https://doi.org/10.1007/978-4-431-55978-8>.
- Backhoff, Julio, Mathias Beiglböck, Yiqing Lin, and Alexander Zalashko. 2020. Causal transport in discrete time and the adapted wasserstein distance. *Probability Theory and Related Fields* 176(3), 967–1006. <https://doi.org/10.1007/s00440-019-00932-w>.
- Brody, Dorje C. and Lane P. Hughston. 2009. Information geometry and asset prices. *International Journal of Theoretical and Applied Finance* 12(2), 135–167. <https://doi.org/10.1142/S0219024909005226>.
- Brunnermeier, M.K. 2009. Deciphering the liquidity and credit crunch 2007–2008. *Journal of Economic Perspectives* 23(1), 77–100.
- Cartea, A., S. Jaimungal, and J. Penalva. 2015. *Algorithmic and High-Frequency Trading*. Cambridge: Cambridge University Press.
- Chen, Y., M. Liu, and Q. Zhang. 2025a. Distributionally robust portfolio optimization with sinkhorn divergences. *Journal of Banking & Finance* 162(1), 107111. <https://doi.org/10.1016/j.jbankfin.2024.107111>.
- Chen, Y., W. Liu, and Q. Zhang. 2025b. Wasserstein distributionally robust portfolio optimisation with regime-switching. *Journal of Banking and Finance* 162, 107234. <https://doi.org/10.1016/j.jbankfin.2025.107234>.
- Cont, Rama. 2001. Empirical properties of asset returns: stylized facts and statistical issues. *Quantitative Finance* 1(2), 223–236. <https://doi.org/10.1080/713665670>.

- Dyrhøberg, Anne Haubo. 2016. Bitcoin, gold and the dollar – a garch volatility analysis. *Finance Research Letters* 16, 85–92. <https://doi.org/10.1016/j.frl.2015.10.008>.
- Guéant, Olivier, Jean-Michel Lasry, and Pierre-Louis Lions. 2012. Optimal execution with nonlinear impact. *SIAM Journal on Financial Mathematics* 3(1), 67–95. <https://doi.org/10.1137/110842186>.
- Haarnoja, Tuomas, Aurick Zhou, Pieter Abbeel, and Sergey Levine. 2018. Soft actor-critic: Off-policy maximum entropy deep reinforcement learning with a stochastic actor. *Proceedings of the 35th International Conference on Machine Learning (ICML) 80*, 1861–1870.
- Hambly, Ben, Renyuan Xu, and Huiling Yang. 2023. Recent advances in reinforcement learning for finance. *Mathematical Finance* 33(2), 345–391. <https://doi.org/10.1111/mafi.12367>.
- Hamilton, James D. 1989. A new approach to the economic analysis of nonstationary time series and the business cycle. *Econometrica* 57(2), 357–384. <https://doi.org/10.2307/1912559>.
- Jarzynski, C. 1997. Nonequilibrium equality for free energy differences. *Physical Review Letters* 78(14), 2690–2693. <https://doi.org/10.1103/PhysRevLett.78.2690>.
- Jiang, Zhengyao and Jinjun Liang. 2017. Deep reinforcement learning for cryptocurrency portfolio management. In *2017 IEEE International Conference on Data Mining Workshops (ICDMW)*, pp. 123–130. <https://doi.org/10.1109/ICDMW.2017.123>.
- Jiang, Z., J. Olmo, and M. Atwi. 2025. High-dimensional multi-period portfolio allocation using deep reinforcement learning. *International Review of Economics and Finance* 92, 103–118. <https://doi.org/10.1016/j.iref.2024.12.005>.
- Kahneman, Daniel and Amos Tversky. 1979. Prospect theory: An analysis of decision under risk. *Econometrica* 47(2), 263–291. <https://doi.org/10.2307/1914185>.
- Kappen, H. J. 2005. Path integrals and symmetry breaking for optimal control theory. *Journal of Statistical Mechanics: Theory and Experiment* 2005(11), P11011. <https://doi.org/10.1088/1742-5468/2005/11/P11011>.
- Kumar, A., V. Singh, and R. Patel. 2026. Geometric deep learning for cryptocurrency portfolio optimisation. *Expert Systems with Applications* 245, 123456. <https://doi.org/10.1016/j.eswa.2025.123456>.
- Ledoit, Olivier and Michael Wolf. 2004. A well-conditioned estimator for large-dimensional covariance matrices. *Journal of Multivariate Analysis* 88(2), 365–411. [https://doi.org/10.1016/S0047-259X\(03\)00096-4](https://doi.org/10.1016/S0047-259X(03)00096-4).
- Levine, Sergey. 2018. Reinforcement learning and control as probabilistic inference: Tutorial and review. *Journal of Machine Learning Research* 19(1), 1–46.
- Liu, Xiao-Yang, Hongyang Yang, Qian Chen, and Weiping Zhang. 2022. Finrl: A deep reinforcement learning library for automated stock trading. *Quantitative Finance* 22(4), 669–683. <https://doi.org/10.1080/14697688.2021.2004314>.
- Liu, Y., Z. Wang, and L. Zhang. 2025. Uncertainty and cryptocurrency volatility: A regime-switching perspective. *Journal of Financial Econometrics* 23(1), 45–78.
- Lo, Andrew W. 2004. The adaptive markets hypothesis: Market efficiency from an evolutionary perspective. *Journal of Portfolio Management* 30(5), 15–29. <https://doi.org/10.3905/jpm.2004.442611>.
- Markowitz, Harry. 1952. Portfolio selection. *The Journal of Finance* 7(1), 77–91. <https://doi.org/10.1111/j.1540-6261.1952.tb01525.x>.
- Metsileng, L.D. and N.D. Moroke. 2026. WOW-E-W paper 2: Navier-Stokes viscosity filtering in cryptocurrency return dynamics. under review.
- Moody, John and Matthew Saffell. 1998. Learning to trade via direct reinforcement. *IEEE Transactions on Neural Networks* 9(5), 875–889. <https://doi.org/10.1109/72.712174>.
- Moroke, N.D. 2026a. CRISPR-DEO: Decision-aware economic dispatch optimization via sparse gradient editing for power system forecasting. *IEEE Access* 14, 31378–31406. <https://doi.org/10.1109/ACCESS.2026.3667829>.
- Moroke, N.D. 2026b. Riemannian geometry and topological data analysis for cryptocurrency liquidity risk: Fisher metrics, persistent homology, and a statistical operations research framework for geodesic execution slippage. ssrn preprint; under review.
- Moroke, N.D. and L.D. Metsileng. 2026. WOW-E-W paper 1: Thermodynamic turbulence in cryptocurrency markets via MS-GARCH-MaxEnt. under review.
- Nystrup, Peter, Benjamin William Hansen, and Henrik Larsen. 2020. Multi-horizon forecasting of portfolio risk using hidden markov models. *Journal of Risk* 22(4), 25–48. <https://doi.org/10.21314/JOR.2020.417>.
- Raffin, Antonin, Ashley Hill, Adam Gleave, Anssi Kanervisto, Maximilian Ernestus, and Noah Dormann. 2021. Stable-baselines3: Reliable reinforcement learning implementations. <https://doi.org/10.5281/zenodo.4667639>.
- Rao, C. Radhakrishna. 1945. Information and the accuracy attainable in the estimation of statistical parameters. *Bulletin of the Calcutta Mathematical Society* 37, 81–91.
- Schulman, J., P. Moritz, S. Levine, M. Jordan, and P. Abbeel. 2016. High-dimensional continuous control using generalised advantage estimation. In *Proceedings of ICLR 2016*.
- Sharpe, William F. 1966. Mutual fund performance. *The Journal of Business* 39(1), 119–138. <https://doi.org/10.1086/294846>.
- van der Vaart, Aad W. 2000. *Asymptotic Statistics*, Volume 3 of *Cambridge Series in Statistical and Probabilistic Mathematics*. Cambridge University Press. <https://doi.org/10.1017/CBO9780511802256>.

- Villani, C. 2003. *Topics in Optimal Transportation*. Providence, RI: American Mathematical Society. 857
- Wang, H., Z. Li, and X. Chen. 2009. Thermodynamic interpretation of market regimes: A maximum entropy approach. *Physica A: Statistical Mechanics and its Applications* 388(18), 3891–3903. <https://doi.org/10.1016/j.physa.2009.05.048>. 858
- Ye, Y., H. Li, and L. Wu. 2020. Reinforcement learning for portfolio management with regime switching. *Expert Systems with Applications* 158, 113548. <https://doi.org/10.1016/j.eswa.2020.113548>. 859
- Zhang, Zihao, Stefan Zohren, and Stephen Roberts. 2020. Deep reinforcement learning for portfolio management: A systematic review. *SSRN Electronic Journal* 1(1), 1–35. <https://doi.org/10.2139/ssrn.3692581>. 860
- Ziebart, Brian D., Andrew Maas, J. Andrew Bagnell, and Anind K. Dey. 2010. Modeling interaction via the principle of maximum causal entropy. In *Proceedings of the 27th International Conference on Machine Learning (ICML)*, pp. 1255–1262. 861